Undergraduate Research Project Presentation

Design and Implementation of a Low-Cost Real-Time Simulator

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Presentation Overview

1. Introduction to CHIL
2. Real-Time Testing Architecture
3. Hardware Interface & Emulation Strategies
4. Simulation & Experiment Results

What is Controller-Hardware-in-the-Loop (CHIL)?

Defining CHIL Simulation

Controller-Hardware-in-the-Loop (CHIL) is a real-time testing methodology where a physical hardware controller is electronically interfaced with a high-fidelity numerical simulation of the power plant executing in real time.

The Cost Barrier in Academia

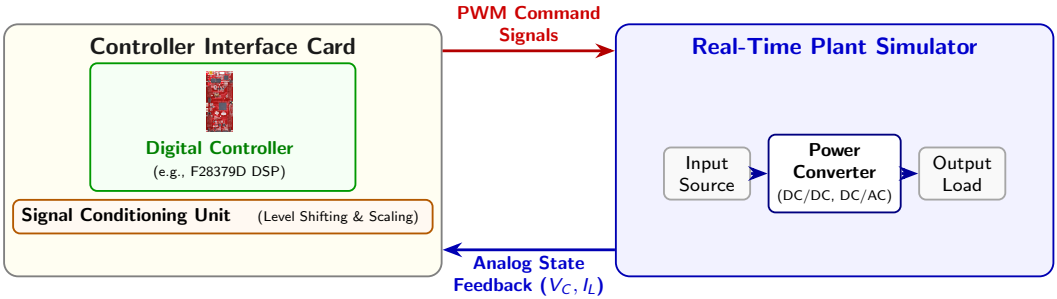
Simulator Platform	Price (USD)	Price (INR)
OPAL-RT (OP4510)	$\sim \$25,000$	∼21,00,000
Typhoon HIL (HIL402)	$\sim \$17,000$	∼14,28,000
dSPACE (MicroLabBox)	$\sim \$15,000$	∼12,60,000
Imperix (BoomBox)	$\sim \$13,000$	∼10,92,000
Plexim (RT-BOX)	$\sim \$10,000$	∼8,40,000
Custom CHIL (v1)	$\sim \$360$	**30,000**
Custom CHIL (v2)	$\sim \$500$	**42,000**

Key Advantages of CHIL:

- Moves beyond pure software (Matlab), by allowing to test hardware controllers with simulated plant systems.
- Huge reduction in risks associated with high-power system, rapid prototyping, quick param changes, safe testing of short circuits

Architecture of the Testing Platform

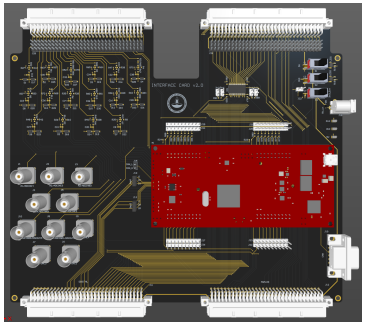
- **The Closed Loop:** The Controller Interface Card executes the control law and issues precise, level-shifted PWM signals → Simulator continuously solves system ODEs → Analog V/I feedback is scaled and read by the controller.



The Controller Interface Card

A Unified Hardware Hub

One important part of this project is the custom-designed **Controller Interface Card** which allows seamless links between controller to any hardware plant configuration. The card functions as a centralized hardware platform, hosting the DSP, optimized signal conditioning circuitry and BNC ports for clean, real-time oscilloscope tracking.



Digital Level Shifting

Safely translates high-speed PWM signals from the onboard controller's logic level (3.3V) to the simulator's logic level without introducing delay.

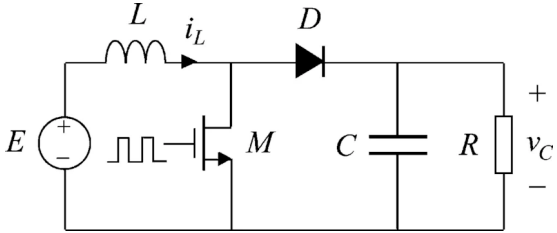
Analog Conditioning

Scales high-fidelity DAC outputs from the simulator down to appropriate ranges for the controller's ADCs, managing offsets and preventing saturation.

Protection & Clamping

Includes transient voltage suppression and clamping circuitry to ensure expensive FPGA/DSP pins are not destroyed during fault emulation.

Mathematical Modeling: The Physical Plant



Ideal Boost Converter Topology

System Formulation

State Vector: $x(t) = \begin{bmatrix} i_L(t) \\ v_C(t) \end{bmatrix}$

Nodal Equations :

Switch ON ($u = 1$)

Inductor charges from source; capacitor discharges into load.

$$\frac{di_L}{dt} = \frac{V_{in}}{L}, \quad \frac{dv_C}{dt} = \frac{-v_C}{R \cdot C}$$

Switch OFF ($u = 0$)

Inductor discharges into the RC network.

$$\frac{di_L}{dt} = \frac{V_{in} - v_C}{L}, \quad \frac{dv_C}{dt} = \frac{i_L}{C} - \frac{v_C}{R \cdot C}$$

Mathematical Modeling: Derived Emulation Models

By mathematically combining the topological states, we derive two distinct models for the real-time simulation platforms:

1. State-Space Averaged Model

Smoothes switching actions over a complete period T_s using duty ratio $d \in [0, 1]$:

$$\frac{d\langle i_L \rangle}{dt} = \frac{V_{in} - (1-d)\langle v_C \rangle}{L}$$

$$\frac{d\langle v_C \rangle}{dt} = \frac{(1-d)\langle i_L \rangle}{C} - \frac{\langle v_C \rangle}{R \cdot C}$$

Type: Continuous time-invariant.

Platform: DSP ($\Delta t \approx 10\,\mu s - 50\,\mu s$).

Fidelity: Tracks fundamental/macroscopic transients; filters high-frequency ripple.

2. High-Frequency Switched Model

Captures discrete topological modifications based on active switch state $u \in \{0, 1\}$:

$$\frac{di_L}{dt} = \frac{V_{in} - (1-u)v_C}{L}$$

$$\frac{dv_C}{dt} = \frac{(1-u)i_L}{C} - \frac{v_C}{R \cdot C}$$

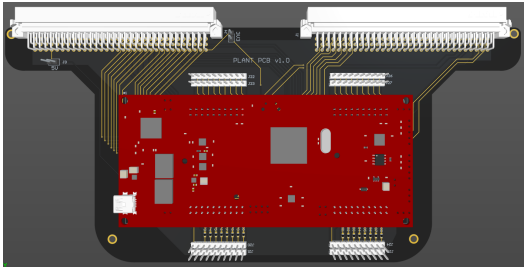
Type: Non-linear piecewise-linear.

Platform: FPGA ($\Delta t \approx 10\,ns - 100\,ns$).

Fidelity: Models exact switching harmonics, precise ripples, and dead-time.

Averaged Plant Model: Results & Parameters

Hardware to Simulate Plant equations :



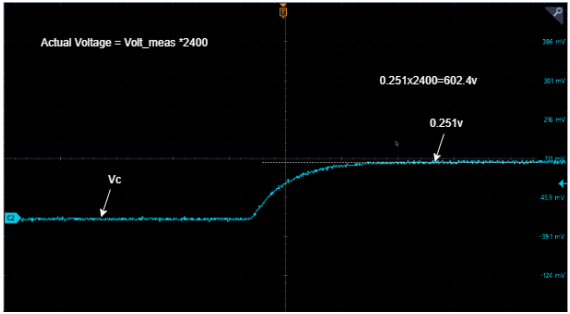
Plant PCB

System Constraints & Expected States

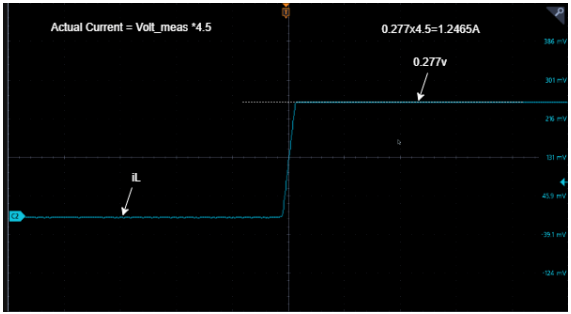
Parameter	Symbol	Value
Design Specifications		
Input Voltage	V_{in}	400.0 V
Switching Frequency	f_{sw}	100 kHz
Duty Cycle	D	33.4%
Component Values		
Inductor	L	600 μH
Capacitor	C	400 μF
Load Resistor	R	720 Ω
Steady-State Output Targets		
Output Voltage	V_C	600.6 V
Inductor Current	I_L	1.246 A
Load Current	I_o	0.83 A

Experimental Validation & Results

Capacitor Voltage



Inductor Current



Switching Model: DSP to FPGA Transition

Why Migrate to FPGA?

- **Parallelism:** i_L and v_C evolve simultaneously in a real circuit. An FPGA updates both in the same clock cycle; a CPU cannot.
- **Intra-cycle Transients:** Since, DSP cannot switch at high frequencies, capturing high frequency ripple with good precision is not a feasible task
- **Altera Max 5** was used to implement the Plant Algorithm.

Ripple & Harmonic Analysis ($D = 0.334$)

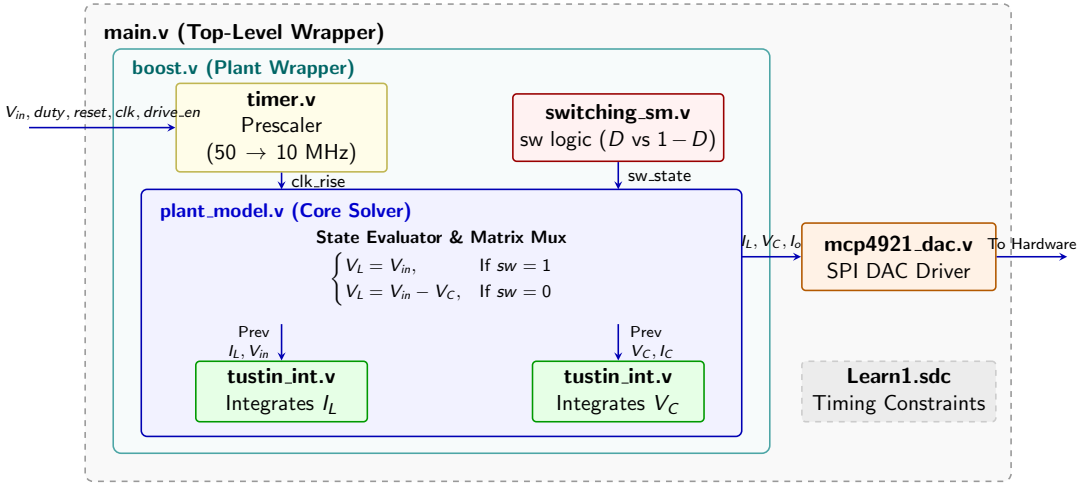
Metric	Analytical Formula	Value
Output Voltage	$V_o = V_{in}/(1-D)$	600.60 V
Load Current	$I_o = V_o/R$	0.83 A
Inductor Ripple	$\Delta i_L = \frac{V_{in}D}{f_{sw}L}$	2.23 A
Voltage Ripple	$\Delta v_C = \frac{I_o D}{f_{sw}C}$	6.97 mV
Low Freq Harmonic	$f_c = \frac{1-D}{2\pi\sqrt{LC}}$	216.4 Hz
Switching Freq.	f_{sw}	100 kHz

Expected Low-Frequency Harmonic

$$f_c = \frac{1-0.334}{2\pi\sqrt{600\mu \cdot 400\mu}} = 216.4 \text{ Hz}$$

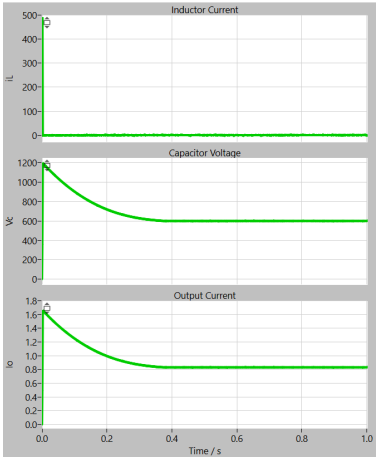
This is the expected effective plant resonance.

FPGA Flow Diagram

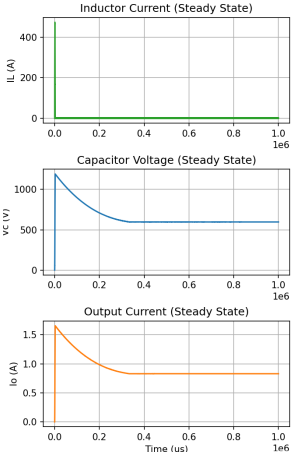


Experimental Validation: Overall Tracking

PLECS Reference



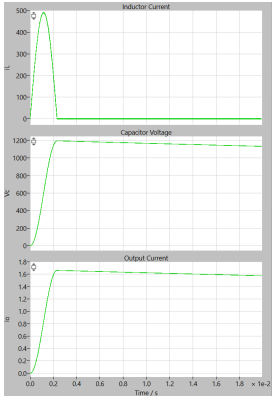
FPGA Real-Time Solver

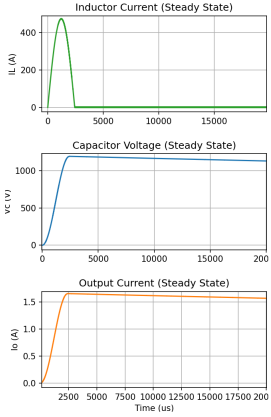


Macroscopic tracking of plant start-up and steady-state operating points.

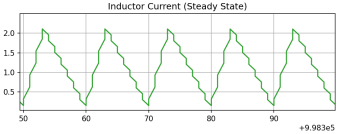
Experimental Validation: Transients & Switching Ripple

Transients zoomed to 0.02 s





Switching Ripple Analysis

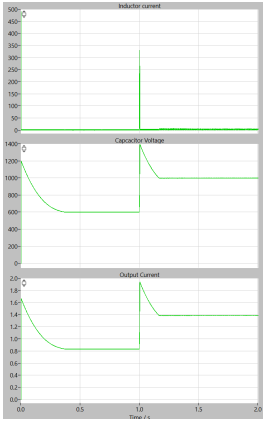


Key Observations

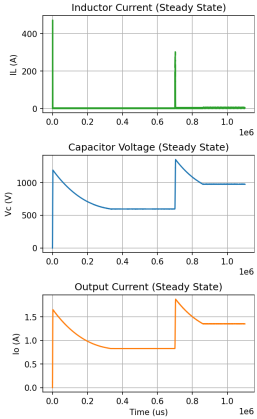
- **Ripple Amplitude:** A steady-state inductor current ripple of $\Delta I_L \approx 2.23$ A and steady state $I_L = 1.24$ A as per design is clearly observed

Transient Validation: Duty Cycle Step Response

PLECS Reference



FPGA Real-Time



Transient Event Data

Pre-Step ($t < 0.7$ s)

- **Duty (D):** 0.334
- **Target V_o:** 600.6 V
- **Natural Frequency (f_n):** 216.4 Hz

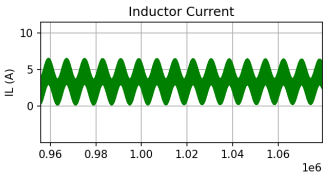
Post-Step ($t \geq 0.7$ s)

- **Duty (D):** **0.600**
- **Target V_o:** **1000.0 V**
- **Natural Frequency(f_n):** **129.9 Hz**

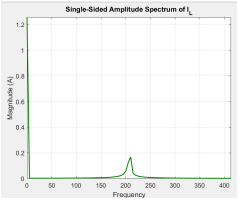
Frequency Domain Analysis: Plant Resonance

Case 1: Duty Cycle, $D = 0.334$

Time-Domain (I_L)



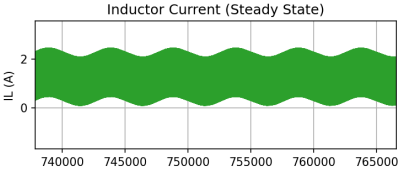
Frequency Spectrum (FFT)



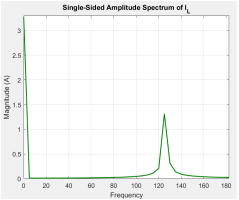
Frequency Peak:
216 Hz

Case 2: Duty Cycle, $D = 0.600$

Time-Domain (I_L)



Frequency Spectrum (FFT)

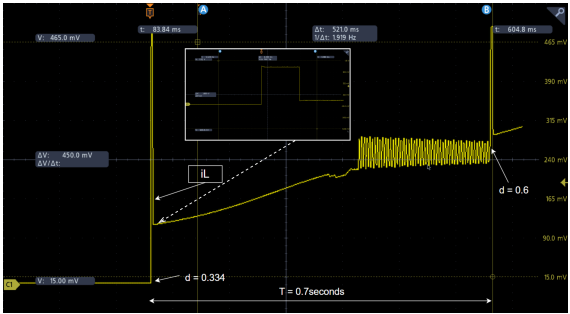


Frequency Peak:
130 Hz

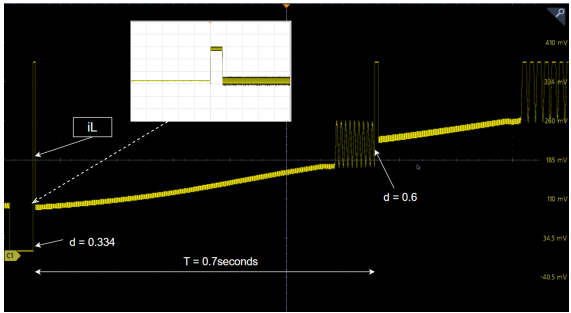
Hardware Implementation Results

Inductor Current:

Typhoon HIL101



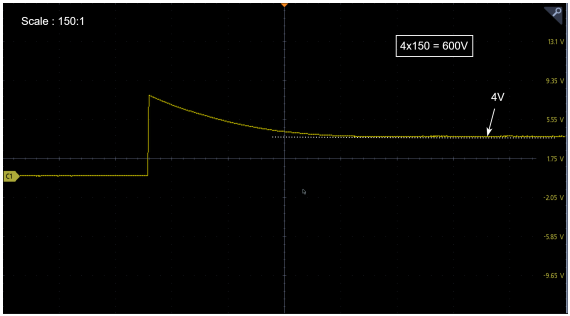
FPGA



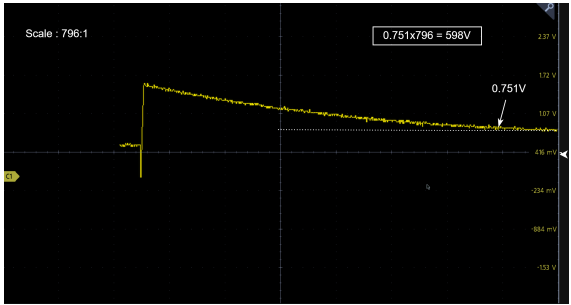
Hardware Implementation Results

Capacitor Voltage:

Typhoon HIL101



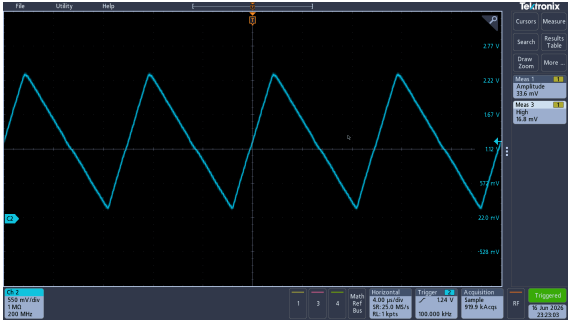
FPGA



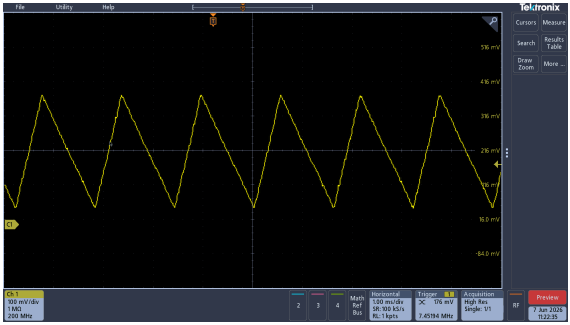
Hardware Implementation Results

Switching Ripple :

Typhoon HIL101



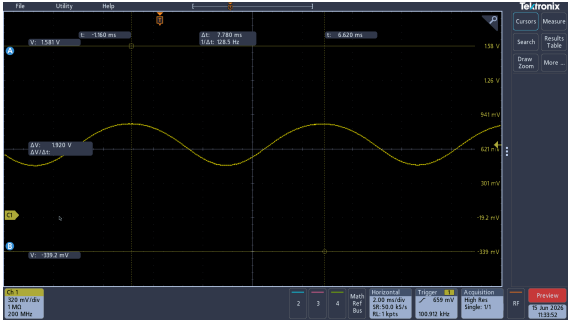
FPGA



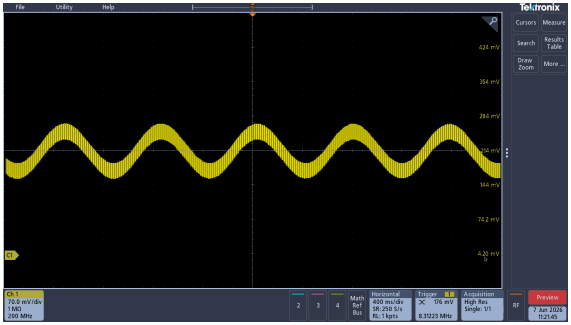
Hardware Implementation Results

Low Frequency Oscillation :

Typhoon HIL101



FPGA



FPGA Implementation: Constraints & Trade-offs

Due to strict hardware limitations on the selected FPGA, several architectural compromises were necessary to synthesize the final design:

- **Logic Element (LE) Deficit:** The full theoretical plant model required 670 LEs, which exceeded the 570 LEs available on the target FPGA.
- **Resource Optimization:** To successfully fit the model onto the silicon, complex calculations had to be approximated and overall numerical precision was reduced.
- **I/O Bandwidth Bottleneck:** While the internal plant solver was optimized to run at a 10MHz frequency, the external 12-bit SPI DAC was physically limited to a 60 kHz update rate.
- **Impact on Analog Observability:** This sampling mismatch resulted in a lot of data buffered at a very slow rate to the DAC resulting in wrong frequencies.

Thank You